

MATH 241

Fall 2017

Problem Set 2

Question 1.

Note that this is fine even if $\lambda=0$

$$\left[\begin{array}{ccc|c} \lambda & 1 & 1 & 0 \\ 1 & \lambda & 1 & 0 \\ 1 & 1 & 1 & 0 \end{array} \right] \xrightarrow{\substack{-\lambda R_3 + R_1 \rightarrow R_1 \\ -R_3 + R_2 \rightarrow R_2}} \left[\begin{array}{ccc|c} 0 & 1-\lambda & 1-\lambda & 0 \\ 0 & \lambda-1 & 0 & 0 \\ 1 & 1 & 1 & 0 \end{array} \right] \xrightarrow{R_2 + R_1 \rightarrow R_1} \left[\begin{array}{ccc|c} 0 & 0 & 1-\lambda & 0 \\ 0 & \lambda-1 & 0 & 0 \\ 1 & 1 & 1 & 0 \end{array} \right]$$

$$\xrightarrow{R_1 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & \lambda-1 & 0 & 0 \\ 0 & 0 & 1-\lambda & 0 \end{array} \right]$$

If $\lambda \neq 1$,

$$\xrightarrow{\substack{\frac{1}{\lambda-1} R_2 \rightarrow R_2 \\ \frac{1}{1-\lambda} R_3 \rightarrow R_3}} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right] \Rightarrow \text{Unique zero solution}$$

If $\lambda = 1$,

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \Rightarrow \text{Non unique solution}$$

Thus, for all $\lambda \neq 1$ there is only the zero solution.

Question 2.

$$\left[\begin{array}{ccc|c} 1 & 2 & -2 & 0 \\ 2 & -1 & \lambda & 0 \\ 3 & 1 & -1 & 0 \end{array} \right] \xrightarrow{\substack{-2R_1 + R_2 \rightarrow R_2 \\ -3R_1 + R_3 \rightarrow R_3}} \left[\begin{array}{ccc|c} 1 & 2 & -2 & 0 \\ 0 & -5 & \lambda+4 & 0 \\ 0 & -5 & 5 & 0 \end{array} \right] \xrightarrow{-R_3 + R_2 \rightarrow R_2} \left[\begin{array}{ccc|c} 1 & 2 & -2 & 0 \\ 0 & 0 & \lambda-1 & 0 \\ 0 & -5 & -5 & 0 \end{array} \right]$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 2 & -2 & 0 \\ 0 & -5 & -5 & 0 \\ 0 & 0 & \lambda-1 & 0 \end{array} \right] \xrightarrow{-\frac{1}{5} R_2 \rightarrow R_2} \left[\begin{array}{ccc|c} 1 & 2 & -2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & \lambda-1 & 0 \end{array} \right]$$

Thus, if $\lambda \neq 1$, the system has only the unique zero solution.

For $\lambda = 1$, the solution is non unique and therefore there are non zero solutions.

Question 3.

$$\left[\begin{array}{ccccc|c} x_1 & x_2 & x_3 & x_4 & x_5 & \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{array} \right] \xrightarrow{-R_1 + R_2 \rightarrow R_2} \left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{array} \right] \xrightarrow{R_3 + R_2 \rightarrow R_2}$$

$$\left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{array} \right] \xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{ccccc|c} \textcircled{x_1} & x_2 & \textcircled{x_3} & \textcircled{x_4} & x_5 & \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{array} \right]$$

Set $x_2 = \alpha$ and $x_5 = \beta$

Then, $x_4 = 0$

$x_3 = -\beta$

$x_1 = -\alpha - \beta$

$$\left. \begin{array}{l} \text{Then, } x_4 = 0 \\ x_3 = -\beta \\ x_1 = -\alpha - \beta \end{array} \right\} \Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -\alpha - \beta \\ \alpha \\ -\beta \\ 0 \\ \beta \end{bmatrix} = \alpha \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \beta \begin{bmatrix} -1 \\ 0 \\ -1 \\ 0 \\ 1 \end{bmatrix} \quad \text{is the general solution for any } \alpha, \beta$$

Question 4.

Let LS_1 be a linear system of equations consisting of (eq. 1), ..., (eq. m) with the solution set S_{LS_1} . Let LS_2 be the linear system of equations with an added equation, (eq. 1), ..., (eq. m), (eq. m+1) with the solution set S_{LS_2} . Since every $s \in S_{LS_2}$ satisfy (eq. 1), ..., (eq. m), (eq. m+1), they all satisfy (eq. 1), ..., (eq. m), therefore a solution of LS_1 . Since every element of S_{LS_2} is also an element of S_{LS_1} , S_{LS_2} is a subset of S_{LS_1} .

Question 5.

It is not always possible. As a trivial example, consider an inconsistent LSE with one equation, $0x = 1$. This cannot be separated into two sets of LSEs, at all. Less trivially, consider $x = 0$

$x = 1$

$x = 2$

which cannot be separated into two sets of LSEs that are both consistent.

Question 6.

It is always possible to reduce a consistent LSE to a unique solution by equating some variables to zero. To see this, consider its reduced row echelon form. Equating the free variables to zero, pivot variables would simply equate to the right hand side, giving the unique solution.